

YASH J. TATE

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EDUCATION

The University of Texas at Austin *B.S in Mechanical Engineering: Business Foundations Minor* **Dec 2027**

GPA: 3.73

Relevant Coursework: Thermodynamics, Mechanics of Solids, Combustion Engine Processes, Engineering Design

Scholarships: Texas Advance Commitment Award, Texas Public Education Grant, University Co-op Material Scholarship

LEADERSHIP EXPERIENCE

Captain & Founder - UT Longhorn Baja Racing (LBR) - Austin, TX **Dec 2024 - Present**

- Co-founded UT Austin's Baja SAE chapter, scaling the organization to 200+ members.
- Directed team operations and recruitment, securing official registration for international SAE competition.

Lead Powertrain Engineer - UT Longhorn Baja Racing (Baja SAE Team) - Austin, TX **April 2025 - Present**

- Designed powertrain components in SolidWorks and authored assembly procedures and DFMEA-informed documentation for powertrain systems.
- Performed confirmation testing to validate countermeasures for 4+ identified failure modes.
- Conducted trade studies that reduced total vehicle mass by ~8 lbs improving vehicle acceleration.

RESEARCH

Undergraduate Researcher - UT SiDi Manufacturing Lab - Austin, TX **Jan 2025 - Present**

- Prototyped and 3D modeled robotic arm end-effectors and print bed components.
- Performed first-order analysis of print path geometry and process parameters, reducing material waste by ~25%.
- Led process documentation and investigated new additive manufacturing technologies to reduce per-part cost.

Undergraduate Researcher - UT Microwave Plasma Generator - Austin, TX **August 2024 - Jan 2025**

- Modeled 10+ dielectric semiconductor geometries in ANSYS HFSS.
- Performed first-order analysis of plasma field uniformity to optimize ammonia synthesis performance.
- Presented findings and developed illustrative geometry comparisons to communicate design tradeoffs.

High School Researcher - Texas State University - San Marcos **Aug 2022 - May 2023**

- Published peer-reviewed research on development and additive manufacturing of thermoplastics bio-based and magnetic composites; presented at SAMPE (2022) and Solid Freeform Fabrication Symposium (2023)

ENGINEERING EXPERIENCE

Baja SAE: Integration Engineer - UT Longhorn Baja Racing (LBR) - Austin, TX **Dec 2024 - April 2025**

- Reduced cycle time by ~30% by creating custom tooling and production aids.
- Improved build repeatability across 3 full reassemblies, reducing build cycle by 4 weeks.
- Led root-cause investigations on 5+ test failures, developing countermeasures and performing confirmation testing.
- Presented root-cause analysis and corrective action, and analytical reports by gathering system-level failure data.

Robot Engineer - Experimental Division - Austin, TX **Aug 2024 - Dec 2024**

- Designed a \$100 combat robot within competition restraints and finished 7th in a 24 team event.
- Investigated failures and implemented countermeasures that improved structural robustness by ~20%.
- Developed test procedures for structural and dynamic loading and presented findings to guide next-cycle design.

TECHNICAL SKILLS

SolidWorks, CATIA V5/V6, Autodesk Fusion 360, MATLAB/Simulink, ANSYS (FEA/CFD), Python, VBA, GD&T (ASME Y14.5), DFMEA/PFMEA, FEA, Manual Fabrication, Jig & Fixture Design, Rapid Prototyping (SLA/FDM), Cross-functional Leadership, Technical Documentation, Vendor Relations, Project Budgeting, Root Cause Analysis

AWARDS

2x Gold Presidents Volunteer Service Award, 1x bronze Presidents Volunteer Award, MotoGP Race Marshall Captain